Curalink

DISEASE

Parkinson's disease

QUERY

gut microbiome treatment

LOCATION

USA

Search

AI SUMMARY

Based on current research, multiple studies suggest promising advancements related to this query. Please refer to the research results below.

RESEARCH RESULTS

PubMed 2026

Gut microbiota modulation via repeated donor fecal transplantation improves motor and gastrointestinal symptoms in drug-naïve Parkinson's disease: a randomized phase 2 trial.

The gut-brain axis is increasingly recognized as a critical contributor to Parkinson's disease (PD) pathogenesis, yet the therapeutic impact of microbiota modulation remains unclear due to lack of clinical trials in drug-naïve patients. We conducted a randomized, double-blind, placebo-controlled phase 2 trial to evaluate the safety, tolerability, and efficacy of repeated donor fecal microbiota transplantation (dFMT) in de novo PD. FMT was administered for seven days (200 mL on days 1-3; 50 mL on days 4-7) per 4-week cycle. Seventy-two patients were randomized 1:1 to receive dFMT or autologous FMT (aFMT), and 66 completed the trial. At 35 weeks, the dFMT group showed significant improvement in motor symptoms (mean change in Unified Parkinson's Disease Rating Scale [UPDRS] III: -3.8 vs. +0.1; $p = 0.0001$) and a substantially greater reduction in constipation severity (dFMT vs. aFMT: -6.5 vs. -0.7; $p < 0.0001$), accompanied by improved quality-

of-life scores. Microbiome profiling revealed greater similarity to donor composition and a marked reduction in Escherichia-Shigella, correlating with decreased colonic α-synuclein aggregation ($r = 0.3775$, $p = 0.0277$), supporting a gut-brain mechanistic link. Biochemical analyses showed elevated fecal dopamine and 3,4-dihydroxyphenylacetic acid levels, while histological assessments demonstrated strengthened epithelial barrier integrity with increased E-cadherin expression. All adverse events were mild and self-limited; no serious treatment-related events were observed. These findings demonstrate that repeated dFMT is safe, well tolerated, and yields clinically meaningful motor and gastrointestinal improvements in drug-naïve PD, providing integrated mechanistic and clinical evidence that microbiota-targeted modulation represents a promising nonpharmacologic therapeutic strategy for neurodegenerative disease. Trial registration: Chinese Clinical Trial Registry, ChiCTR2200064151.

Authors: Rui Zhang, Renyi Feng, Jiuqi Wang, Yongkang Chen, Han Liu, Qingyong Zhu, Haiyan Tian, Chi Qin, Junfang Teng, Beisha Tang, Min Wu, Jinsheng Zeng, Erxi Wu, Xuebing Ding, Xuejing Wang

PubMed 2025

Supplementation with short-chain fatty acids and a prebiotic improves clinical outcome in Parkinson's disease: a randomized double-blind prospective study.

Parkinson's disease is associated with a dysbiotic, proinflammatory gut microbiome, disruptions to intestinal barrier functions, and immunological imbalance. Microbiota-produced short-chain fatty acids, such as propionic and butyric acid promote gut barrier integrity and immune regulation, but their impact on Parkinson's disease pathology remains mostly unknown. In a randomized double-blind prospective study, 72 people with Parkinson's disease received propionic and butyric acid and/or the prebiotic fiber 2'-fucosyllactose supplementation over 6 months in combination with existing Parkinson's disease-specific therapy. Patients underwent complete neurological assessment and provided blood and stool samples before as well as 3 and 6 months after supplementation. We observed a robust improvement in motor symptoms, with all intervention groups achieving clinically meaningful reductions. These motor benefits were paralleled by clinically relevant reductions in levodopa medication. In contrast, effects on nonmotor symptoms were more heterogeneous. Notably, the interventions also modulated peripheral immune responses and enhanced mitochondrial respiration in immunocytes. Postintervention microbiota remodeled inflammatory and barrier-related gene sets in gut organ cultures and improved in vitro barrier functions. Treatment response was associated with microbiome composition, distinct patterns of colonic transcription and permeability ex vivo. Multiobjective analysis revealed immune parameters associated with an optimal response to supplementation. Short-chain fatty acids ameliorate clinical symptoms in Parkinson's disease patients and modulate intestinal and peripheral immunity. This clinical trial was retrospectively registered with the German Clinical Trials Register (DRKS), registration number DRKS00027061 on 11/19/2021.

Authors: Tobias Hegelmaier, Alexander Duscha, Christiane Desel, Sabrina Fuchs, Michal Shapira, Sivan Amidror, Qihao Shan, Gabriele I Stangl, Frank Hirche, Stefan Kempa, András Maifeld, Lisa-Marie Würtele, Jana Peplinski, Diana Jauk, Gitali Naim, Nuphar Shidlovsky, Adva Cohen, Yifat Bennet, Lisa

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PubMed 2025

Changes in Gut Microbiome Following Acupuncture and Moxibustion in Patients With Parkinson Disease: Protocol for a Single-Group, Prospective, Observational Study.

Parkinson disease (PD), a prevalent neurodegenerative disorder characterized by motor and nonmotor symptoms, is becoming increasingly prevalent worldwide. Conventional treatment for PD involves dopamine therapy, including levodopa; however, this treatment is ineffective for nonmotor symptoms and may cause adverse effects. The gut-brain axis has been hypothesized to promote PD, and regulation of gut microbiome, which modulates the gut-brain axis, is emerging as a treatment target. Acupuncture and moxibustion exert therapeutic effects on PD and modulate the gut microbial composition. We present a protocol for analyzing the effects of acupuncture and moxibustion on gut microbiome and exploring its association with symptoms in patients with PD. This single-group, prospective, observational study will recruit 60 patients with idiopathic PD and 20 healthy participants. Baseline gut microbiome patterns and motor and nonmotor symptoms of both groups will be compared. Patients with PD will be treated with acupuncture, moxibustion, and intradermal acupuncture twice a week for 12 weeks (24 sessions total). Motor and nonmotor symptoms and gut microbiome changes in patients with PD will be compared before starting treatment (day 0), during treatment (6 weeks), at the end of treatment (12 weeks), and 2 months after the end of treatment (20 weeks). The correlation between motor and nonmotor symptoms of PD changed by acupuncture and moxibustion treatment and changes in gut microbiome will be analyzed. Healthy participants will be assessed for motor and nonmotor symptoms of PD and gut microbiome after screening. This study was supported by the National Research Foundation of Korea, funded by the Ministry of Science and ICT (Information and Communication Technology), Republic of Korea, and recruitment for the study started on October 21, 2021. As of February 19, 2025, recruitment and observation ended, and data analysis is being conducted. This is the first clinical study to assess the effects of acupuncture and moxibustion on gut microbiome and explore its association with symptoms in patients with PD. The results will provide clinical evidence to explain the microbiome-gut-brain axis mechanism of acupuncture and moxibustion for PD and suggest the possibility of acupuncture as an alternative therapy for PD. Clinical Research Information Service KCT0006669; https://tinyurl.com/42jsxs5a. DERR1-10.2196/76551.

Authors: Han-Gyul Lee, Seungwon Kwon, Mijung Yeom, Sun-Young Bae, Hi-Joon Park

PubMed 2024

Safety and efficacy of faecal microbiota transplantation in patients with mild to moderate Parkinson's disease (GUT-PARFECT): a double-blind, placebo-controlled,

randomised, phase 2 trial.

Dysregulation of the gut microbiome has been implicated in Parkinson's disease (PD). This study aimed to evaluate the clinical effects and safety of a single faecal microbiota transplantation (FMT) in patients with early-stage PD. The GUT-PARFECT trial, a single-centre randomised, double-blind, placebo-controlled trial was conducted at Ghent University Hospital between December 01, 2020 and December 12, 2022. Participants (aged 50-65 years, Hoehn and Yahr stage 2) were randomly assigned to receive nasojejunal FMT with either healthy donor stool or their own stool. Computer-generated randomisation was done in a 1:1 ratio through permutated-block scheduling. Treatment allocation was concealed for participants and investigators. The primary outcome measure at 12 months was the change in the Movement Disorders Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) motor score obtained during off-medication evaluations. Intention-to-treat analysis was performed using a mixed model for repeated measures analysis. This completed trial is registered on ClinicalTrials.gov (NCT03808389). Between December 2020 and December 2021, FMT procedures were conducted on 46 patients with PD: 22 in the healthy donor group and 24 in the placebo group. Clinical evaluations were performed at baseline, 3, 6, and 12 months post-FMT. Full data analysis was possible for 21 participants in the healthy donor group and 22 in the placebo group. After 12 months, the MDS-UPDRS motor score significantly improved by a mean of 5.8 points (95% CI -11.4 to -0.2) in the healthy donor group and by 2.7 points (-8.3 to 2.9) in the placebo group (p = 0.0235). Adverse events were limited to temporary abdominal discomfort. Our findings suggested a single FMT induced mild, but long-lasting beneficial effects on motor symptoms in patients with early-stage PD. These findings highlight the potential of modulating the gut microbiome as a therapeutic approach and warrant a further exploration of FMT in larger cohorts of patients with PD in various disease stages. Flemish PD patient organizations (VPL and Parkili), Research Foundation Flanders (FWO), Biocodex Microbiota Foundation.

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PubMed 2024

Clearing the Smoke: What Protects Smokers from Parkinson's Disease?

The link between smoking and a lower risk of Parkinson's disease (PD) is one of the strongest environmental or lifestyle associations in neuroepidemiology. Growing evidence supports the hypothesis that the association is based on a neuroprotective effect of smoking on PD, despite the plausible alternative that smoking serves as a marker for a proximal protective influence without itself conferring benefit. But how smoking could protect against neurodegeneration in PD is not well understood. Of several candidate molecules and mechanisms that have been nominated, nicotine has received the most attention. However, randomized controlled clinical trials of nicotine in PD have failed to demonstrate benefit on motor endpoints, including the NIC-PD study in which recently diagnosed participants were randomly assigned to placebo or nicotine treatment for 1 year. Given these results, the time is right to evaluate the neuroprotective

potential of other molecules and biochemical cascades triggered by smoking. Here, we review the evidence supporting smoking's possible protective effect on PD, compounds in tobacco and smoke that might mediate such benefit, and non-causal classes of explanation, including reverse causation and the prospect of shared genetic determinants of smoking and PD resistance. The therapeutic potential of non-nicotine components of smoke is suggested by studies supporting multiple alternative mechanisms ranging from monoamine oxidase inhibitors to gut microbiome disruption to antioxidant response induction by chronic exposure to low levels of carbon monoxide. Rigorous investigation is warranted to evaluate this molecule and others for disease-preventing and disease-modifying activity in PD models and, if warranted, in clinical trials. © 2024 The Authors. Movement Disorders published by Wiley Periodicals LLC on behalf of International Parkinson and Movement Disorder Society.

Authors: Kenneth N Rose, Michael A Schwarzschild, Stephen N Gomperts

PubMed 2023

Protocol for a multicentre cross-sectional, longitudinal ambulatory clinical trial in rheumatoid arthritis and Parkinson's disease patients analysing the relation between the gut microbiome, fasting and immune status in Germany (ExpoBiome).

Chronic inflammatory diseases like rheumatoid arthritis (RA) and neurodegenerative disorders like Parkinson's disease (PD) have recently been associated with a decreased diversity in the gut microbiome, emerging as key driver of various diseases. The specific interactions between gut-borne microorganisms and host pathophysiology remain largely unclear. The microbiome can be modulated by interventions comprising nutrition.The aim of our clinical study is to (1) examine effects of prolonged fasting (PF) and time-restricted eating (TRE) on the outcome parameters and the immunophenotypes of RA and PD with (2) special consideration of microbial taxa and molecules associated with changes expected in (1), and (3) identify factors impacting the disease course and treatment by in-depth screening of microorganisms and molecules in personalised HuMiX gut-on-chip models, to identify novel targets for anti-inflammatory therapy. This trial is an open-label, multicentre, controlled clinical trial consisting of a cross-sectional and a longitudinal study. A total of 180 patients is recruited. For the cross-sectional study, 60 patients with PD, 60 patients with RA and 60 healthy controls are recruited at two different, specialised clinical sites. For the longitudinal part, 30 patients with PD and 30 patients with RA undergo 5-7 days of PF followed by TRE (16:8) for a period of 12 months. One baseline visit takes place before the PF intervention and 10 follow-up visits will follow over a period of 12 months (April 2021 to November 2023). Ethical approval was obtained to plan and conduct the trial from the institutional review board of the Charité-Universitätsmedizin Berlin (EA1/204/19), the ethics committee of the state medical association (Landesärztekammer) of Hessen (2021-2230-zvBO) and the Ethics Review Panel (ERP) of the University of Luxembourg (ERP 21-001 A ExpoBiome). The results of this study will be disseminated through peer-reviewed publications, scientific presentations and social media. NCT04847011.

Authors: Bérénice Hansen, Cédric C Laczny, Velma T E Aho, Audrey Frachet-Bour, Janine Habier, Marek Ostaszewski, Andreas Michalsen, Etienne Hanslian, Daniela A Koppold, Anika M Hartmann,

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PubMed 2023

Effect of strain Shirota supplementation on clinical responses and gut microbiome in Parkinson's disease.

Parkinson's disease (PD) is a common neurodegenerative disease characterized by motor issues and a range of non-motor symptoms. Microbial therapy may be a useful approach for the treatment of PD. However, comprehensive analyses of the impact of probiotic supplementation on motor and non-motor symptoms are still lacking and the mechanisms whereby the treatment works remain unclear. This study investigated strain Shirota (LcS) supplementation on clinical responses, gut microbiota and faecal metabolites in PD patients. Patients (= 128) were randomised to receive either probiotics (LcS-fermented milk, containing 1 × 10 living LcS cells) or placebo for 12 weeks. All participants were examined and the basic clinical features were recorded using questionnaires. Fecal and blood samples were collected at the baseline and after 12 weeks for further omics analysis. We found that LcS intervention significantly alleviated patients' constipation-related symptoms and non-motor symptoms. We found no significant shifts in the composition of gut microbiota or faecal metabolites. Several taxa were differentially abundant between the groups, especially with regard to LcS intake, which increased the abundance of the genus in the probiotic group compared with those at the baseline and in the placebo group. The faecal concentration of L-tyrosine was significantly decreased and the plasma concentration of L-tyrosine was increased in the probiotic group compared with the placebo group. Our study demonstrated that although supplementation with LcS did not induce major changes in the global gut microbiome, the probiotic had favorable effects in managing constipation and other non-motor symptoms in PD patients. This study was registered at the Chinese Clinical Trial Registry: ChiCTR1800016795.
Authors: Xiaodong Yang, Xiaoqin He, Shaoqing Xu, Yi Zhang, Chengjun Mo, Yiqiu Lai, Yanyan Song, Zheng Yan, Penghui Ai, Yiwei Qian, Qin Xiao

PubMed 2023

Fecal microbiota transplantation in Parkinson's disease-A randomized repeat-dose, placebo-controlled clinical pilot study.

The intestinal microbiome plays a primary role in the pathogenesis of neurodegenerative disorders and may provide an opportunity for disease modification. We performed a pilot clinical study looking at the safety of fecal microbiota transplantation (FMT), its effect on the microbiome, and improvement of symptoms in Parkinson's disease. This was a randomized, double-blind placebo-controlled pilot study, wherein orally administered lyophilized FMT product or matching placebo was given to 12 subjects with mild to moderate Parkinson's disease with constipation twice weekly for 12 weeks. Subjects were followed for safety and clinical

improvement for 9 additional months (total study duration 12 months). Fecal microbiota transplantation caused non-severe transient upper gastrointestinal symptoms. One subject receiving FMT was diagnosed with unrelated metastatic cancer and was removed from the trial. Beta diversity (taxa) of the microbiome, was similar comparing placebo and FMT groups at baseline, however, for subjects randomized to FMT, it increased significantly at 6 weeks (= 0.008) and 13 weeks (= 0.0008). After treatment with FMT, proportions of selective families within the phylum Firmicutes increased significantly, while proportion of microbiota belonging to Proteobacteria were significantly reduced. Objective motor findings showed only temporary improvement while subjective symptom improvements were reported compared to baseline in the group receiving FMT. Constipation, gut transient times (NS), and gut motility index (= 0.0374) were improved in the FMT group. Subjects with Parkinson's disease tolerated multi-dose-FMT, and experienced increased diversity of the intestinal microbiome that was associated with reduction in constipation and improved gut transit and intestinal motility. Fecal microbiota transplantation administration improved subjective motor and non-motor symptoms. ClinicalTrial.gov, identifier: NCT03671785.

Authors: Herbert L DuPont, Jessika Suescun, Zhi-Dong Jiang, Eric L Brown, Heather T Essigmann, Ashley S Alexander, Andrew W DuPont, Tehseen Iqbal, Netanya S Utay, Michael Newmark, Mya C Schiess

CLINICAL TRIALS

N/A

No trials available

Location: N/A
Eligibility: N/A